ELITE CYBERSECURITY SPECIALIST · MALAYSIA

BEYOND CONVENTIONAL CIS LEVEL 2 BASELINES

KHAHP — Kicksecure High-Assurance Hardening Project
Full Technical Evidence & Engineering Register

More than finding exploits.
Reduce exposure. Deepen isolation.
Verify defenses. Preserve recovery.

44	308	924
Hardening domains	Control families	A/B/C checkpoints

An evidence-led examination of existing Kicksecure defenses, applied configuration improvements, additional defense-in-depth opportunities, controlled negative testing, and persistence/recovery requirements.

v4.1 · EDITORIAL REVISION · 19 OCTOBER 2026

Evidence cutoff: 18 October 2026. This revision does not assert additional machine tests, independently certified benchmarks, or finalized 924-gate acceptance. Detailed results and acceptance boundaries follow.

Public Consolidated Release v4.1 · 19 October 2026 (editorial update)
Elite Cybersecurity Specialist · Malaysia
200+ hours of owner-reported engineering · 44 sections · 308 control families · 924 A/B/C assurance checkpoints
Technical observations and outstanding decisions: through 18 October 2026

> **Engineering mission:** Go beyond a conventional baseline-only hardening exercise. Extend Kicksecure's existing defenses by identifying security configuration weaknesses, evaluating additional protective controls, reducing unnecessary attack surface and producing repeatable evidence of enforcement, persistence and recovery — whether or not a specific weakness has been exploited.

1. Executive engineering position and integrity statement

KHAHP is a high-assurance defensive hardening and verification program for a Kicksecure / Debian 13 laptop. It is not a vulnerability-scanner scorecard. Its core objective is to **maximize practical defensive strength and operating-system resilience**, while respecting vendor safeguards, USER/SYSMAINT privilege separation and physical recovery.

1.1. What "beyond CIS Level 2" means in this project

The engineering program takes **applicable CIS Debian Linux 13 Level 2 Workstation** recommendations as a reference point and investigates further isolation, kernel and boot controls, network and DNS path validation, device/privacy boundaries, supply-chain assurance, recovery and evidence-driven negative testing. **DISA General Purpose Operating System SRG / applicable STIG-inspired objectives**, **KSPP**, and Kicksecure-native safeguards are additional design references.

The term *beyond a conventional benchmark-only baseline* describes the **breadth and intended depth of the engineering program**, not a third-party-certified ranking against every bank or operating-system deployment. A complete control-by-control comparison of applicable CIS/DISA requirements, native equivalents and project extensions is the explicit work of Section 43; this release does not invent a benchmark score or official certification.

A **security weakness does not need a working exploit** to merit correction. Overly permissive isolation, unnecessary accessible services, weak privacy defaults, missing detection capability or untested failure recovery can represent practical weaknesses, and hardening can deliberately exceed the minimum required by a checklist. Conversely, a planned check or optional feature alone is not proof that stock Kicksecure is defective.

1.2. Scope and significance of the 924 checkpoints

Program measure	Number	Engineering significance
Hardening sections	**44**	Defense domains, including boot, kernel, filesystem, network, devices, monitoring and recovery
Technical control families	**308**	Seven specific control families per section (`01.01` to `44.07`)
A/B/C assurance checkpoints	**924**	Per-family provenance (A), runtime enforcement (B), persistence/recovery (C)
Historical mixed backlog entries	**69**	Mixed review, manual, exception and implementation records, not 69 confirmed vulnerabilities
Historical 01–38 action records	**63**	Tracked follow-up decisions, not automatically 63 pending patches
Distinct supported change events	**At least 3**	Traceable corrective/hardening examples in Section 3, not a complete lifetime fix count
Reviewer-signed A/B/C gates	**0** in the supplied 18 Oct ledger	Historical formal signature record awaiting evidence reconciliation; not 924 failed controls

Program measure	Number	Engineering significance
Planned commercial-image tests	**60**	Future clean-image, ISO, KVM/QCOW2 and OVA release tests

The **924 entries are meaningful security assurance checkpoints**: each can reveal a configuration problem, confirm an existing safeguard, evaluate a defense-in-depth enhancement, or prove compatibility and survival through system lifecycle events. They **must not be flattened into either 924 discovered exploits or 924 meaningless paperwork entries**.

1.3. Demonstrated defensive engineering and improvement opportunities

Domain	Supported engineering position	Deeper assurance / improvement being pursued
USER / SYSMAINT separation	Distinct roles and privileged maintenance workflow demonstrated	Complete linked role and recovery acceptance without enabling USER sudo
Kernel and memory	Observed restricted ptrace, kernel pointers, dmesg, BPF, ASLR and protected-file controls	Link configuration provenance to kernel build, effective coverage and update regression
Storage / filesystem	LUKS root observed; `/tmp` hardened with `nodev,nosuid`; boot partition mount ordering corrected and immediately validated	Recovery, boot persistence and compatibility verification
Software confinement	AppArmor broadly enabled and browser subprocess seccomp observed	Firefox vendor placeholder remains `unconfined`: a concrete candidate for safer restrictive confinement design/testing
Resolver / network	Local Unbound and DNSSEC negative/positive behavior tested; configured DoT, nftables and Tor observed	Negative proof of encrypted upstream trust, resolver bypass prevention, firewall packet verdicts and application routing
Peripheral / privacy	USBGuard device rules, USER camera/microphone open denial, Bluetooth block and thumbnail restrictions observed	Physical, hotplug and power/lid persistence tests under authorized owner policy
Audit / integrity	AIDE daily timer observed	Section 40 audit event collection, rules and AIDE alert lifecycle still require end-to-end validation

Provenance matters: Existing vendor protections are credited as Kicksecure/Debian capabilities where supported; *project-specific changes* are identified separately. Configuration evidence can be technically meaningful before formal acceptance, but a proposed mitigation is not described as applied unless supported by a dated change/result.

1.4. Security finding categories and acceptance boundaries

Classification	Operational meaning
Observed protection	Installed/configured or tested safeguard, with its actual observed scope explicitly recorded
Confirmed security weakness / defect	Demonstrated undesirable behavior or misconfiguration against an applicable approved security requirement
Hardening enhancement opportunity	An additional reduction in exposure or improvement in isolation/visibility/resilience, including when an exploit has not been reproduced
Policy/compatibility decision	Intentional choice, exception, N/A, or restriction with potential functional/recovery trade-offs
Evidence pending / review	Provenance, runtime behavior, negative test or persistence not yet sufficiently evidenced for the intended claim
Formally accepted	Approved A/B/C outcome linked to evidence, date and authorized reviewer or recorded exception

Do not equate **not exploitable** with **not worth hardening**. Do not equate **not yet tested** with **confirmed insecure**. The same controls may have proven individual aspects and still await full A/B/C review.

1.5. Project status at this editorial release

- **Active engineering milestone:** Sections 01–40 are within the owner-reported implementation/investigation workstream. This does not mean all their acceptance gates are signed.
- **Section 39:** BIOS/firmware is owner-declared complete and expressly excluded from retesting; respect the decision and do not fabricate independent signatures.
- **Section 40:** Auditd/AIDE end-to-end event coverage, alert lifecycle, rule enforcement and retention are active completion tasks.
- **Sections 41–44:** Planned final assurance, recovery, standards mapping and formal acceptance phases.
- **Evidence ledger:** The supplied 18 October CSV records 0/924 formal reviewer signatures. Earlier successful observations and changes remain valid within their stated test scope; all 924 gates have not been demonstrated as security failures.
- **Revision boundary:** v4.1 updates explanatory language, organization and presentation only. It adds **no newly executed machine tests, remediations or certification results** beyond the cited October 18 records.

2. Evidence rules and status vocabulary

Gate	Minimum evidence	Failure to prove does not imply
A — provenance	Package / configuration owner, original and changed value, intended policy, approved diff	No security feature exists
B — effective enforcement	Exact role, real-world allowed behavior, bounded denied behavior and observed outcomes	A test was passed merely because a unit was active
C — persistence and recovery	Relevant reboot, reconnect, update or isolated recovery plus rollback where required	A freshly observed setting will persist indefinitely

Statuses used in this public release: `OBSERVED`, `TEST-PASS`, `OWNER-COMPLETE`, `FIXED-IMMEDIATE-VERIFIED`, `REVIEW`, `MANUAL-REQUIRED`, `EXCEPTION`, `NOT-TESTED`, and `FORMALLY-ACCEPTED`. A source finding may be sound without being formally accepted. Do not treat `REVIEW` as `CONFIRMED-VULNERABILITY`. Formal acceptance requires an authorized reviewer, date and linked A/B/C proof.

The supplied CSV contained 924 `OPEN` gates. This edition **preserves that historical signature state** rather than counterfeiting signatures from command output. The private updated master supplies newer evidence candidates for owner/reviewer reconciliation.

3. Configuration changes, corrective actions and durable results

ID	Section	Change and observed result	Remaining boundary
CHG-07-01	07	`/tmp`: `nodev,nosuid` applied; prior USER reboot persistence observed	`noexec` intentionally excluded for compatibility; other mounts require own decisions
CHG-37-01	37	GNOME thumbnail generation disabled (`false` → `true`); later synthetic/reboot and PCManFM-Qt checks observed	Other applications, deletion policy and format-specific viewers require separate review
KHAHP-20261018-FSTAB-ORDER	07	`/boot` entry moved before `/boot/efi`; root `findmnt --verify --verbose`: no errors/warnings; backup retained	Validate after next operator-controlled reboot; change is not evidence of previous exploit

Not changed on purpose

- No blanket `/tmp noexec`; application and installer compatibility are explicit considerations.
- No sudden USER privilege escalation or weakening of SYSMAINT separation for testing convenience.
- No unreviewed global firewall/DNS policy cutover, Tor stop, network manager teardown or blanket USBGuard deny-all.
- No AIDE database reinitialization or invasive camera/microphone capture.
- Section 39 firmware/BIOS policy remains **owner-completed and excluded from retesting**.

4. High-value verification observations and limitations

Domain	Later dated technical evidence	Interpretation
OS/platform	Debian 13 trixie; Linux 6.12.111+deb13-amd64; zero failed systemd units in collected runs	Observed role- and time-scoped baseline
Kernel	`kptr_restrict=2`, `dmesg_restrict=1`, `ptrace_scope=3`, unprivileged BPF restricted, ASLR and protected file knobs observed	Effective values for inspected controls, not all kernel options proven
Secure Boot / modules	Secure Boot enabled in SYSMAINT snapshot; signed Tirdad build provenance observed	Signed-module rejection and future update behavior not yet signed off
Network resolver	Unbound listening on loopback, active, enabled and `unbound-checkconf` clean	Local resolver availability does not rule out app-specific DNS bypass
DNSSEC	Positive AD response and broken-DNSSEC SERVFAIL confirmed	DNS answer authenticity in bounded test; no transport confidentiality inference
DNS-over-TLS	Upstream Quad9 forwarding configuration, authenticated-name and TLS port 853 observations previously recorded	Negative certificate-path validation and all egress routes require independent test
nftables	Active rules; input/forward default-drop; OUTPUT policy accepts with explicit external direct DNS drops	Rules present, external UDP/TCP53 packet verdicts not fully signed
Tor	Tor active; SOCKS loopback listener present	Which apps are Tor-scoped and fail-closed not established by service activity
USBGuard	Device-specific authorized mouse and approved recovery/storage; no destructive policy changes	Keyboards/mice/storage allowed by owner policy; actual input and composite behavior require physical proof
Camera / mic	USER open() denied for two video nodes and one microphone capture node; no recording	Current nonprivileged direct-node test only, no guarantee against all privileged/hardware paths
Bluetooth	rfkill soft blocks on both adapters and inactive service	Current block state observed
Browser	Firefox 158 active; `network.trr.mode=5`, WebRTC/prefetch restrictions and seccomp-filtered subprocesses observed	`/etc/apparmor.d/firefox` intentionally unconfined; no strict MAC claim
Desktop privacy	Thumbnail disabling retained; synthetic test and reboot persistence recorded; `swaylock` lock/unlock worked	Five-minute and lid/resume tests not passed by short lock test
Audit/AIDE	`dailyaidecheck.timer` enabled/active. SYSMAINT `auditctl -s`: audit enabled, PID 0; `auditctl -l`: no rules	Do not declare full audit-event or integrity alert lifecycle accepted

5. Implementation review by all 44 sections

Each section below records the **strongest supported later observation**, an improvement or acceptance boundary, and seven control families with exact A/B/C identifiers. Together these 308 families form the **security configuration and high-assurance evidence surface** of the project. They include inherited protections, applied project hardening, candidate security improvements and outstanding tests. The listed gate IDs identify acceptance requirements; their existence does not alone indicate a flaw or a completed signature.

01 — Platform / OS / Hardware Baseline

Observed implementation/evidence: Debian 13 (trixie), kernel 6.12.111+deb13-amd64 and zero failed systemd units were observed in the October 18 USER/SYSMAINT snapshots.

Acceptance boundary / next proof: Reconcile full hardware inventory and post-upgrade provenance to formal A/B/C acceptance.

Family	Exact technical scope	A/B/C evidence IDs
01.01	Operating-system identity and Debian/ Kicksecure release baseline	KHAHP-01.01A · KHAHP-01.01B · KHAHP-01.01C
01.02	Kernel release, architecture and build identity	KHAHP-01.02A · KHAHP-01.02B · KHAHP-01.02C
01.03	CPU model, topology and virtualization capability	KHAHP-01.03A · KHAHP-01.03B · KHAHP-01.03C
01.04	Physical memory and swap capacity baseline	KHAHP-01.04A · KHAHP-01.04B · KHAHP-01.04C
01.05	Storage controller, disk and filesystem topology	KHAHP-01.05A · KHAHP-01.05B · KHAHP-01.05C
01.06	UEFI/bare-metal/virtualization platform state	KHAHP-01.06A · KHAHP-01.06B · KHAHP-01.06C
01.07	Failed-unit, uptime, boot-time and baseline health state	KHAHP-01.07A · KHAHP-01.07B · KHAHP-01.07C

02 — Kicksecure Package Provenance

Observed implementation/evidence: The latest `dpkg --audit` returned no pending package-database problems. Provenance must be derived from real owning packages, not package-name guesses.

Acceptance boundary / next proof: Attribute outstanding permission-hardener policy and package-origin discrepancies.

Historical action references: `U01-003`.

Family	Exact technical scope	A/B/C evidence IDs
02.01	Core Kicksecure package installation and exact versions	KHAHP-02.01A · KHAHP-02.01B · KHAHP-02.01C
02.02	Supporting security-package installation and exact versions	KHAHP-02.02A · KHAHP-02.02B · KHAHP-02.02C
02.03	APT source files and repository configuration	KHAHP-02.03A · KHAHP-02.03B · KHAHP-02.03C
02.04	APT policy and repository origin attribution	KHAHP-02.04A · KHAHP-02.04B · KHAHP-02.04C
02.05	Package ownership of security-critical configuration files	KHAHP-02.05A · KHAHP-02.05B · KHAHP-02.05C

Family	Exact technical scope	A/B/C evidence IDs
02.06	Package holds, partial installs and dpkg database health	KHAHP-02.06A · KHAHP-02.06B · KHAHP-02.06C
02.07	Local modifications versus package-owned baseline	KHAHP-02.07A · KHAHP-02.07B · KHAHP-02.07C

03 — USER / SYSMAINT Privilege Architecture

Observed implementation/evidence: The intended USER/SYSMAINT privilege separation was retained during testing; privileged maintenance ran as UID 0 in a separate maintenance context.

Acceptance boundary / next proof: Cross-reference authoritative boot-role markers and regression checks; no USER sudo enablement.

Family	Exact technical scope	A/B/C evidence IDs
03.01	USER, SYSMAINT and root account separation	KHAHP-03.01A · KHAHP-03.01B · KHAHP-03.01C
03.02	Distinct UID/GID and home-directory boundaries	KHAHP-03.02A · KHAHP-03.02B · KHAHP-03.02C
03.03	user-sysmaint-split package and owned policy files	KHAHP-03.03A · KHAHP-03.03B · KHAHP-03.03C
03.04	sudo binary ownership, mode and SYSMAINT-only execution	KHAHP-03.04A · KHAHP-03.04B · KHAHP-03.04C
03.05	sudoers syntax and user-sysmaint- split policy	KHAHP-03.05A · KHAHP-03.05B · KHAHP-03.05C
03.06	USER negative sudo/root acquisition probe	KHAHP-03.06A · KHAHP-03.06B · KHAHP-03.06C
03.07	Login-shell and boot-role separation behavior	KHAHP-03.07A · KHAHP-03.07B · KHAHP-03.07C

04 — SYSMAINT Boot Role

Observed implementation/evidence: SYSMAINT evidence was collected in a privileged maintenance session; normal USER boot retained ordinary browser and network functionality.

Acceptance boundary / next proof: Tie protected role markers, role-specific units and provenance to current release acceptance.

Family	Exact technical scope	A/B/C evidence IDs
04.01	boot-role=sysmaint kernel marker	KHAHP-04.01A · KHAHP-04.01B · KHAHP-04.01C
04.02	systemd.unit=sysmaint-boot.target marker	KHAHP-04.02A · KHAHP-04.02B · KHAHP-04.02C
04.03	sysmaint-boot.target package provenance and unit content	KHAHP-04.03A · KHAHP-04.03B · KHAHP-04.03C
04.04	Required SYSMAINT dependencies and wanted units	KHAHP-04.04A · KHAHP-04.04B · KHAHP-04.04C
04.05	Intentionally omitted USER-only or incompatible hardeners	KHAHP-04.05A · KHAHP-04.05B · KHAHP-04.05C
04.06	Conflicting USER/graphical boot-role negative checks	KHAHP-04.06A · KHAHP-04.06B · KHAHP-04.06C

Family	Exact technical scope	A/B/C evidence IDs
04.07	Failed units and SYSMAINT boot health	KHAHP-04.07A · KHAHP-04.07B · KHAHP-04.07C

05 — Accounts / UID / GID / Local Identity

Observed implementation/evidence: Restricted USER account and service-account inventory were reviewed; no arbitrary account changes were justified.

Acceptance boundary / next proof: Document the `sync` service-account classification and identity persistence.

Historical action references: `U01-004`.

Family	Exact technical scope	A/B/C evidence IDs
05.01	root, USER and SYSMAINT account identity	KHAHP-05.01A · KHAHP-05.01B · KHAHP-05.01C
05.02	Duplicate UID and duplicate GID detection	KHAHP-05.02A · KHAHP-05.02B · KHAHP-05.02C
05.03	UID 0 exclusivity and privileged- account re- view	KHAHP-05.03A · KHAHP-05.03B · KHAHP-05.03C
05.04	Low-UID service-account shell classification	KHAHP-05.04A · KHAHP-05.04B · KHAHP-05.04C
05.05	Home-directory ownership and permissions	KHAHP-05.05A · KHAHP-05.05B · KHAHP-05.05C
05.06	shadow/gshadow/passwd/group ownership and modes	KHAHP-05.06A · KHAHP-05.06B · KHAHP-05.06C
05.07	NSS identity source configuration and root lock state	KHAHP-05.07A · KHAHP-05.07B · KHAHP-05.07C

06 — LUKS / Disk / Swap Encryption

Observed implementation/evidence: LUKS-encrypted root block mapping was observed; the removable installation/ recovery disk was kept separate from system storage.

Acceptance boundary / next proof: Private key/header backup, encrypted backup and isolated restore demonstration remain separate acceptance requirements.

Historical action references: `U01-009`.

Family	Exact technical scope	A/B/C evidence IDs
06.01	Root filesystem source and dm-crypt mapping	KHAHP-06.01A · KHAHP-06.01B · KHAHP-06.01C
06.02	LUKS2 backing partition and cryptsetup status	KHAHP-06.02A · KHAHP-06.02B · KHAHP-06.02C
06.03	LUKS header metadata without key- material disclosure	KHAHP-06.03A · KHAHP-06.03B · KHAHP-06.03C
06.04	Root encryption chain from block device to mountpoint	KHAHP-06.04A · KHAHP-06.04B · KHAHP-06.04C
06.05	Active swap inventory and encrypted place- ment	KHAHP-06.05A · KHAHP-06.05B · KHAHP-06.05C
06.06	swapfile ownership, mode and containing filesystem	KHAHP-06.06A · KHAHP-06.06B · KHAHP-06.06C

Family	Exact technical scope	A/B/C evidence IDs
06.07	crypttab/fstab encryption persistence and boot mapping	KHAHP-06.07A · KHAHP-06.07B · KHAHP-06.07C

07 — Filesystem / Mount Security

Observed implementation/evidence: A controlled earlier change set `/tmp` to `nodev,nosuid`. On October 18 the `/boot` entry was placed before `/boot/efi`; privileged `findmnt --verify --verbose` reported no warnings or errors, with a timestamped backup. `noexec` remains deliberately excluded.

Acceptance boundary / next proof: Capture next approved post-reboot mount-order persistence and application compatibility; do not apply global noexec merely to match a benchmark.

Historical action references: `U01-001`, `U01-002`, `U01-011`, `U01-012`, `U01-014`.

Family	Exact technical scope	A/B/C evidence IDs
07.01	Effective mount table and filesystem inventory	KHAHP-07.01A · KHAHP-07.01B · KHAHP-07.01C
07.02	Root mount source, filesystem and options	KHAHP-07.02A · KHAHP-07.02B · KHAHP-07.02C
07.03	/boot and /boot/efi separation, ownership and exposure	KHAHP-07.03A · KHAHP-07.03B · KHAHP-07.03C
07.04	/tmp tmpfs, sticky bit and mount- option review	KHAHP-07.04A · KHAHP-07.04B · KHAHP-07.04C
07.05	/var/tmp sticky-bit and separate- mount review	KHAHP-07.05A · KHAHP-07.05B · KHAHP-07.05C
07.06	/home containment, ownership and inherited options	KHAHP-07.06A · KHAHP-07.06B · KHAHP-07.06C
07.07	Writable mountpoints, fstab and remount-secure review	KHAHP-07.07A · KHAHP-07.07B · KHAHP-07.07C

08 — GRUB / Bootloader Protection

Observed implementation/evidence: GRUB protection directives were observed in the historical technical review.

Acceptance boundary / next proof: Physical unauthorized edit-denial and protected recovery proof have not been supplied; no secret disclosure.

Historical action references: `U01-008`.

Family	Exact technical scope	A/B/C evidence IDs
08.01	GRUB package baseline and generated grub.cfg	KHAHP-08.01A · KHAHP-08.01B · KHAHP-08.01C
08.02	grub.cfg ownership, permissions and menu entries	KHAHP-08.02A · KHAHP-08.02B · KHAHP-08.02C
08.03	GRUB password_pbkdf2 and superusers directives	KHAHP-08.03A · KHAHP-08.03B · KHAHP-08.03C
08.04	Kicksecure CPU/kernel/ram-wipe/remount/signed -module fragments	KHAHP-08.04A · KHAHP-08.04B · KHAHP-08.04C
08.05	Quiet-boot and recovery-restriction fragments	KHAHP-08.05A · KHAHP-08.05B · KHAHP-08.05C

Family	Exact technical scope	A/B/C evidence IDs
08.06	Effective boot command-line correlation	KHAHP-08.06A · KHAHP-08.06B · KHAHP-08.06C
08.07	/boot and EFI exposure versus cryptographic integrity limits	KHAHP-08.07A · KHAHP-08.07B · KHAHP-08.07C

09 — CPU Vulnerability Mitigations

Observed implementation/evidence: Spectre-class, SMT and memory-mitigation kernel options were observed; boot parameters alone are not proof of every mitigation.

Acceptance boundary / next proof: Correlate kernel capability, microcode and runtime vulnerabilities rather than counting cmdline tokens.

Family	Exact technical scope	A/B/C evidence IDs
09.01	CPU identity, vendor and mitigation- capability baseline	KHAHP-09.01A · KHAHP-09.01B · KHAHP-09.01C
09.02	SMT/Hyper-Threading disablement and topo- logy	KHAHP-09.02A · KHAHP-09.02B · KHAHP-09.02C
09.03	Runtime microcode revision and package baseline	KHAHP-09.03A · KHAHP-09.03B · KHAHP-09.03C
09.04	Kernel vulnerability-interface enumeration	KHAHP-09.04A · KHAHP-09.04B · KHAHP-09.04C
09.05	Spectre/Meltdown/L1TF/MDS/ Retbleed/BHI/ SSB/TAA/GDS mitigations	KHAHP-09.05A · KHAHP-09.05B · KHAHP-09.05C
09.06	Mitigation-related kernel command- line para- meters	KHAHP-09.06A · KHAHP-09.06B · KHAHP-09.06C
09.07	CPU mitigation flags and kernel-log confirma- tion	KHAHP-09.07A · KHAHP-09.07B · KHAHP-09.07C

10 — IOMMU / DMA Protection

Observed implementation/evidence: The SYSMAINT evidence reported 17 IOMMU groups and enabled Secure Boot.

Acceptance boundary / next proof: Firmware/preboot DMA applicability remains a separate hardware/recovery question; owner BIOS work is not reopened.

Historical action references: `U01-010`.

Family	Exact technical scope	A/B/C evidence IDs
10.01	CPU virtualization/VT-d capability baseline	KHAHP-10.01A · KHAHP-10.01B · KHAHP-10.01C
10.02	IOMMU kernel parameters and passthrough policy	KHAHP-10.02A · KHAHP-10.02B · KHAHP-10.02C
10.03	DMAR/IOMMU runtime enablement	KHAHP-10.03A · KHAHP-10.03B · KHAHP-10.03C
10.04	Translated default domain and strict invalida- tion	KHAHP-10.04A · KHAHP-10.04B · KHAHP-10.04C
10.05	IRQ/interrupt remapping	KHAHP-10.05A · KHAHP-10.05B · KHAHP-10.05C

Family	Exact technical scope	A/B/C evidence IDs
10.06	IOMMU group topology and full PCI membership	KHAHP-10.06A · KHAHP-10.06B · KHAHP-10.06C
10.07	Early PCI DMA restriction and firmware WSMT evidence	KHAHP-10.07A · KHAHP-10.07B · KHAHP-10.07C

11 — Kernel Sysctl Hardening

Observed implementation/evidence: `kernel.kptr_restrict=2`, `dmesg_restrict=1`, `ptrace_scope=3`, unprivileged BPF restrictions and protected file operations were observed.

Acceptance boundary / next proof: Verify package precedence and post-upgrade persistence where material.

Family	Exact technical scope	A/B/C evidence IDs
11.01	Sysctl source-tree inventory and precedence	KHAHP-11.01A · KHAHP-11.01B · KHAHP-11.01C
11.02	Kicksecure security-misc sysctl source files	KHAHP-11.02A · KHAHP-11.02B · KHAHP-11.02C
11.03	Kernel information-leak restrictions	KHAHP-11.03A · KHAHP-11.03B · KHAHP-11.03C
11.04	ASLR, mmap, kexec and privileged- coredump controls	KHAHP-11.04A · KHAHP-11.04B · KHAHP-11.04C
11.05	BPF, ptrace, perf and TTY line- discipline restrictions	KHAHP-11.05A · KHAHP-11.05B · KHAHP-11.05C
11.06	Protected links/FIFOs/regular-file controls	KHAHP-11.06A · KHAHP-11.06B · KHAHP-11.06C
11.07	Network-stack sysctl hardening observations	KHAHP-11.07A · KHAHP-11.07B · KHAHP-11.07C

12 — Kernel Boot / Memory Hardening

Observed implementation/evidence: Memory and allocator hardening boot flags were observed in the effective command line.

Acceptance boundary / next proof: Do not promote CFI/KCFI command-line tokens into confirmed kernel enforcement without build/runtime proof.

Historical action references: `U01-005`.

Family	Exact technical scope	A/B/C evidence IDs
12.01	Slab allocator hardening and slab_nomerge	KHAHP-12.01A · KHAHP-12.01B · KHAHP-12.01C
12.02	init_on_alloc and init_on_free memory initialization	KHAHP-12.02A · KHAHP-12.02B · KHAHP-12.02C
12.03	page allocator shuffle and stack randomization	KHAHP-12.03A · KHAHP-12.03B · KHAHP-12.03C
12.04	vsyscall, vdso32 and 32-bit compatibility restrictions	KHAHP-12.04A · KHAHP-12.04B · KHAHP-12.04C
12.05	debugfs, KFENCE and kernel debug- surface controls	KHAHP-12.05A · KHAHP-12.05B · KHAHP-12.05C

Family	Exact technical scope	A/B/C evidence IDs
12.06	CFI/KCFI and boot-time control acceptance	KHAHP-12.06A · KHAHP-12.06B · KHAHP-12.06C
12.07	pstore/ERST/proc_mem/entropy trust boot parameters	KHAHP-12.07A · KHAHP-12.07B · KHAHP-12.07C

13 — Kernel Module Restrictions

Observed implementation/evidence: Kernel/module policy inventories exist from earlier paired-role observations.

Acceptance boundary / next proof: Resolve built-in versus loadable module applicability without loading prohibited test modules.

Historical action references: U01-006.

Family	Exact technical scope	A/B/C evidence IDs
13.01	security-misc module blacklist configuration	KHAHP-13.01A · KHAHP-13.01B · KHAHP-13.01C
13.02	Hard-disable install rules for risky modules	KHAHP-13.02A · KHAHP-13.02B · KHAHP-13.02C
13.03	Loaded-module inventory versus policy	KHAHP-13.03A · KHAHP-13.03B · KHAHP-13.03C
13.04	Legacy filesystem module restrictions	KHAHP-13.04A · KHAHP-13.04B · KHAHP-13.04C
13.05	Legacy network/protocol module restrictions	KHAHP-13.05A · KHAHP-13.05B · KHAHP-13.05C
13.06	USB/storage/media module restriction exceptions	KHAHP-13.06A · KHAHP-13.06B · KHAHP-13.06C
13.07	Module-load negative tests and compatibility exceptions	KHAHP-13.07A · KHAHP-13.07B · KHAHP-13.07C

14 — Kernel Module Signing / DKMS

Observed implementation/evidence: Privileged evidence identified signed Tirdad DKMS builds; Secure Boot was enabled in the October 18 SYSMAINT check.

Acceptance boundary / next proof: Signed-module rejection, future kernel-update regression and key-provenance acceptance require controlled proof.

Historical action references: U01-007, U01-013.

Family	Exact technical scope	A/B/C evidence IDs
14.01	Loaded-module signer inventory	KHAHP-14.01A · KHAHP-14.01B · KHAHP-14.01C
14.02	DKMS package and built-module inventory	KHAHP-14.02A · KHAHP-14.02B · KHAHP-14.02C
14.03	tirdad DKMS build, signer and runtime state	KHAHP-14.03A · KHAHP-14.03B · KHAHP-14.03C
14.04	NVIDIA/other third-party module signer state	KHAHP-14.04A · KHAHP-14.04B · KHAHP-14.04C
14.05	Kernel module signature enforcement capability	KHAHP-14.05A · KHAHP-14.05B · KHAHP-14.05C

Family	Exact technical scope	A/B/C evidence IDs
14.06	MOK/enrolled certificate readiness	KHAHP-14.06A · KHAHP-14.06B · KHAHP-14.06C
14.07	Secure-Boot compatibility and unsigned-module exceptions	KHAHP-14.07A · KHAHP-14.07B · KHAHP-14.07C

15 — AppArmor / Mandatory Access Control

Observed implementation/evidence: AppArmor is enabled with many loaded profiles; Firefox 158 uses a vendor-owned `flags=(unconfined)` placeholder, and USER browser processes were observed `firefox (unconfined)`. Selected Firefox processes did have seccomp filters.

Acceptance boundary / next proof: Full restrictive Firefox confinement is not implemented: prepare an isolated compatible profile and test complain/enforce transitions with rollback; do not `aa-enforce` the placeholder.

Historical action references: `U15-001`, `U15-002`, `U15-003`.

Family	Exact technical scope	A/B/C evidence IDs
15.01	AppArmor kernel/module enablement	KHAHP-15.01A · KHAHP-15.01B · KHAHP-15.01C
15.02	Profile inventory and parser status	KHAHP-15.02A · KHAHP-15.02B · KHAHP-15.02C
15.03	Enforce/complain/unconfined process counts	KHAHP-15.03A · KHAHP-15.03B · KHAHP-15.03C
15.04	Tor and network-service profile enforcement	KHAHP-15.04A · KHAHP-15.04B · KHAHP-15.04C
15.05	Browser/application profile coverage	KHAHP-15.05A · KHAHP-15.05B · KHAHP-15.05C
15.06	Denied-event review and false-positive handling	KHAHP-15.06A · KHAHP-15.06B · KHAHP-15.06C
15.07	USER versus SYSMAINT AppArmor behavior	KHAHP-15.07A · KHAHP-15.07B · KHAHP-15.07C

16 — SUID / SGID / File Capabilities

Observed implementation/evidence: SUID/SGID and capabilities were inventoried. No evidence justifies blanket removal of package-owned helper capabilities.

Acceptance boundary / next proof: Resolve privileged-binary drift and capability-specific exception decisions.

Historical action references: `U15-004`, `U15-005`.

Family	Exact technical scope	A/B/C evidence IDs
16.01	Complete SUID file inventory	KHAHP-16.01A · KHAHP-16.01B · KHAHP-16.01C
16.02	Complete SGID file inventory	KHAHP-16.02A · KHAHP-16.02B · KHAHP-16.02C
16.03	Package ownership of privileged binaries	KHAHP-16.03A · KHAHP-16.03B · KHAHP-16.03C
16.04	permission-hardener expected mode validation	KHAHP-16.04A · KHAHP-16.04B · KHAHP-16.04C

Family	Exact technical scope	A/B/C evidence IDs
16.05	File-capability inventory and package attribution	KHAHP-16.05A · KHAHP-16.05B · KHAHP-16.05C
16.06	Unexpected privileged executable detection	KHAHP-16.06A · KHAHP-16.06B · KHAHP-16.06C
16.07	Privilege-binary negative/compatibility review	KHAHP-16.07A · KHAHP-16.07B · KHAHP-16.07C

17 — Permissions / ACL / Umask

Observed implementation/evidence: Restrictive file- and temporary-directory controls were observed. A vendor writable-status exception is documented.

Acceptance boundary / next proof: Accept or reject vendor-permission exceptions on evidence; test post-update drift.

Historical action references: `U15-006`, `U15-007`.

Family	Exact technical scope	A/B/C evidence IDs
17.01	System file ownership baseline	KHAHP-17.01A · KHAHP-17.01B · KHAHP-17.01C
17.02	World-writable system path inventory	KHAHP-17.02A · KHAHP-17.02B · KHAHP-17.02C
17.03	Unexpected group-writable sensitive paths	KHAHP-17.03A · KHAHP-17.03B · KHAHP-17.03C
17.04	ACL inventory on sensitive configuration	KHAHP-17.04A · KHAHP-17.04B · KHAHP-17.04C
17.05	Root and user umask policy	KHAHP-17.05A · KHAHP-17.05B · KHAHP-17.05C
17.06	Home and evidence-directory permissions	KHAHP-17.06A · KHAHP-17.06B · KHAHP-17.06C
17.07	Sticky-bit and shared-directory permission policy	KHAHP-17.07A · KHAHP-17.07B · KHAHP-17.07C

18 — PAM / Login / Faillock

Observed implementation/evidence: Yescrypt/PAM hardening was observed. The SYSMAINT review found `faillock deny=50` and `unlock_time=never`.

Acceptance boundary / next proof: Document safe lockout recovery and approved password/aging choices without deliberate account lockout.

Historical action references: `U15-008`, `U15-009`, `U15-010`.

Family	Exact technical scope	A/B/C evidence IDs
18.01	PAM stack package provenance	KHAHP-18.01A · KHAHP-18.01B · KHAHP-18.01C
18.02	password quality and hashing policy	KHAHP-18.02A · KHAHP-18.02B · KHAHP-18.02C
18.03	faillock/brute-force lockout policy	KHAHP-18.03A · KHAHP-18.03B · KHAHP-18.03C
18.04	login delay and retry controls	KHAHP-18.04A · KHAHP-18.04B · KHAHP-18.04C

Family	Exact technical scope	A/B/C evidence IDs
18.05	su/sudo/polkit PAM interaction	KHAHP-18.05A · KHAHP-18.05B · KHAHP-18.05C
18.06	account aging and inactive-account policy	KHAHP-18.06A · KHAHP-18.06B · KHAHP-18.06C
18.07	USER/SYSMAINT login regression tests	KHAHP-18.07A · KHAHP-18.07B · KHAHP-18.07C

19 — SSH Client / Server Security

Observed implementation/evidence: The latest USER snapshot showed the SSH server inactive and no TCP SSH listener; local client and socket uses are classified separately.

Acceptance boundary / next proof: Confirm listener/role persistence after update without enabling a network SSH daemon.

Historical action references: `U15-011`, `U15-012`.

Family	Exact technical scope	A/B/C evidence IDs
19.01	Local SSH server service/socket state	KHAHP-19.01A · KHAHP-19.01B · KHAHP-19.01C
19.02	sshd listener and inbound port absence	KHAHP-19.02A · KHAHP-19.02B · KHAHP-19.02C
19.03	sshd_config security-misc policy if installed	KHAHP-19.03A · KHAHP-19.03B · KHAHP-19.03C
19.04	SSH client configuration and host-key verification	KHAHP-19.04A · KHAHP-19.04B · KHAHP-19.04C
19.05	Private-key ownership and permission baseline	KHAHP-19.05A · KHAHP-19.05B · KHAHP-19.05C
19.06	Password-authentication and root-login policy for future server use	KHAHP-19.06A · KHAHP-19.06B · KHAHP-19.06C
19.07	Known-hosts/key algorithm and crypto-policy review	KHAHP-19.07A · KHAHP-19.07B · KHAHP-19.07C

20 — Enabled / Running Service Audit

Observed implementation/evidence: Two October 18 health checks found zero failed systemd units.

Acceptance boundary / next proof: Attribute expected services per USER/SYSMAINT role; monitor post-upgrade changes rather than disable vendor units indiscriminately.

Historical action references: `U15-013`, `U15-014`.

Family	Exact technical scope	A/B/C evidence IDs
20.01	Enabled systemd service inventory	KHAHP-20.01A · KHAHP-20.01B · KHAHP-20.01C
20.02	Running systemd service inventory	KHAHP-20.02A · KHAHP-20.02B · KHAHP-20.02C
20.03	Socket/path/timer activated service inventory	KHAHP-20.03A · KHAHP-20.03B · KHAHP-20.03C
20.04	Kicksecure-required service attribution	KHAHP-20.04A · KHAHP-20.04B · KHAHP-20.04C

Family	Exact technical scope	A/B/C evidence IDs
20.05	Unexpected or unnecessary service classification	KHAHP-20.05A · KHAHP-20.05B · KHAHP-20.05C
20.06	Listener-to-service ownership correlation	KHAHP-20.06A · KHAHP-20.06B · KHAHP-20.06C
20.07	Reboot persistence and failed-service review	KHAHP-20.07A · KHAHP-20.07B · KHAHP-20.07C

21 — systemd Service Sandboxing

Observed implementation/evidence: Existing systemd sandboxing is present; exposure scores are not themselves exploitable vulnerabilities.

Acceptance boundary / next proof: Evaluate any optional resolver restrictions in a test build with DNS and boot-role regression.

Historical action references: U15-015, U15-016.

Family	Exact technical scope	A/B/C evidence IDs
21.01	systemd-analyze security baseline	KHAHP-21.01A · KHAHP-21.01B · KHAHP-21.01C
21.02	NoNewPrivileges adoption	KHAHP-21.02A · KHAHP-21.02B · KHAHP-21.02C
21.03	ProtectSystem/ProtectHome restrictions	KHAHP-21.03A · KHAHP-21.03B · KHAHP-21.03C
21.04	PrivateTmp/PrivateDevices isolation	KHAHP-21.04A · KHAHP-21.04B · KHAHP-21.04C
21.05	CapabilityBoundingSet and ambient capabilities	KHAHP-21.05A · KHAHP-21.05B · KHAHP-21.05C
21.06	RestrictNamespaces/SystemCallFilter/address-family controls	KHAHP-21.06A · KHAHP-21.06B · KHAHP-21.06C
21.07	Service-specific compatibility exceptions and regressions	KHAHP-21.07A · KHAHP-21.07B · KHAHP-21.07C

22 — System.map / Core Dumps / PStore

Observed implementation/evidence: Crash-dump/pstore suppression controls and associated privacy-versus-forensics tradeoffs were documented.

Acceptance boundary / next proof: Record exception rationale and observe package/kernel update drift; do not generate a production kernel panic.

Historical action references: U15-017, U15-018.

Family	Exact technical scope	A/B/C evidence IDs
22.01	System.map exposure and removal policy	KHAHP-22.01A · KHAHP-22.01B · KHAHP-22.01C
22.02	systemd-coredump configuration	KHAHP-22.02A · KHAHP-22.02B · KHAHP-22.02C
22.03	kernel.core_pattern effective state	KHAHP-22.03A · KHAHP-22.03B · KHAHP-22.03C

Family	Exact technical scope	A/B/C evidence IDs
22.04	fs.suid_dumpable effective state	KHAHP-22.04A · KHAHP-22.04B · KHAHP-22.04C
22.05	pstore retention/disable policy	KHAHP-22.05A · KHAHP-22.05B · KHAHP-22.05C
22.06	EFI pstore and ERST boot controls	KHAHP-22.06A · KHAHP-22.06B · KHAHP-22.06C
22.07	Crash-evidence privacy versus diagnostic requirements	KHAHP-22.07A · KHAHP-22.07B · KHAHP-22.07C

23 — Journald / Logging / Logrotate

Observed implementation/evidence: Journald/secure-time baseline checks exist. Logging integrity is not equivalent to complete audit rule coverage.

Acceptance boundary / next proof: Approve retention and forwarding rules, ensure time coherence and protect evidence custody.

Historical action references: `U15-019`, `U15-020`.

Family	Exact technical scope	A/B/C evidence IDs
23.01	journald security-misc configuration	KHAHP-23.01A · KHAHP-23.01B · KHAHP-23.01C
23.02	Persistent versus volatile journal storage	KHAHP-23.02A · KHAHP-23.02B · KHAHP-23.02C
23.03	Journal file permissions and ownership	KHAHP-23.03A · KHAHP-23.03B · KHAHP-23.03C
23.04	Rate limiting and size/retention policy	KHAHP-23.04A · KHAHP-23.04B · KHAHP-23.04C
23.05	Forwarding/syslog integration state	KHAHP-23.05A · KHAHP-23.05B · KHAHP-23.05C
23.06	logrotate configuration and sensitive-log modes	KHAHP-23.06A · KHAHP-23.06B · KHAHP-23.06C
23.07	Log integrity, time coherence and reboot persistence	KHAHP-23.07A · KHAHP-23.07B · KHAHP-23.07C

24 — Panic / Recovery / Emergency Shutdown

Observed implementation/evidence: Emergency/recovery restrictions and shutdown service behavior were inventoried.

Acceptance boundary / next proof: Use disposable media/VM for fault injection; do not trigger an unexpected live-system panic.

Historical action references: `U15-021`, `U15-022`.

Family	Exact technical scope	A/B/C evidence IDs
24.01	kernel panic-on-oops policy	KHAHP-24.01A · KHAHP-24.01B · KHAHP-24.01C
24.02	panic timeout and fail-closed behavior	KHAHP-24.02A · KHAHP-24.02B · KHAHP-24.02C
24.03	emerg-shutdown.service provenance and state	KHAHP-24.03A · KHAHP-24.03B · KHAHP-24.03C

Family	Exact technical scope	A/B/C evidence IDs
24.04	rescue/emergency target restrictions	KHAHP-24.04A · KHAHP-24.04B · KHAHP-24.04C
24.05	rd.emergency=halt and rd.shell=0 behavior	KHAHP-24.05A · KHAHP-24.05B · KHAHP-24.05C
24.06	Magic SysRq and recovery-surface policy	KHAHP-24.06A · KHAHP-24.06B · KHAHP-24.06C
24.07	Controlled failure/recovery evidence and roll-back	KHAHP-24.07A · KHAHP-24.07B · KHAHP-24.07C

25 — RAM Wipe

Observed implementation/evidence: Kernel memory initialization controls and encrypted storage mapping are observed.

Acceptance boundary / next proof: Verify wipe-hook lifecycle and swap/crypt ancestry without claiming immunity to cold-boot attacks.

Historical action references: U25-003, U25-004, U25-017.

Family	Exact technical scope	A/B/C evidence IDs
25.01	ram-wipe package provenance and version	KHAHP-25.01A · KHAHP-25.01B · KHAHP-25.01C
25.02	GRUB/package RAM-wipe source presence	KHAHP-25.02A · KHAHP-25.02B · KHAHP-25.02C
25.03	init_on_alloc/init_on_free live correlation	KHAHP-25.03A · KHAHP-25.03B · KHAHP-25.03C
25.04	Shutdown/reboot wipe hook inventory	KHAHP-25.04A · KHAHP-25.04B · KHAHP-25.04C
25.05	Swap/LUKS interaction during shutdown	KHAHP-25.05A · KHAHP-25.05B · KHAHP-25.05C
25.06	Controlled reboot/shutdown persistence test	KHAHP-25.06A · KHAHP-25.06B · KHAHP-25.06C
25.07	Limitations of software RAM wiping and cold-boot scope	KHAHP-25.07A · KHAHP-25.07B · KHAHP-25.07C

26 — USBGuard / Peripheral Security

Observed implementation/evidence: USBGuard was active and enabled; its policy authorized a recognized ZOWIE mouse and SanDisk removable medium. Approved USB mice, keyboards and recovery storage are allowed; a webcam was USB-authorized but its USER video device nodes denied direct access.

Acceptance boundary / next proof: Second mouse, external-keyboard functional acceptance, phone/network composite negative testing and policy precedence still need controlled proof.

Historical action references: U25-001, U25-002, U25-018.

Family	Exact technical scope	A/B/C evidence IDs
26.01	usbguard daemon and DBus service state	KHAHP-26.01A · KHAHP-26.01B · KHAHP-26.01C
26.02	Policy files and rule provenance	KHAHP-26.02A · KHAHP-26.02B · KHAHP-26.02C

Family	Exact technical scope	A/B/C evidence IDs
26.03	ImplicitPolicyTarget and device-policy settings	KHAHP-26.03A · KHAHP-26.03B · KHAHP-26.03C
26.04	Allowed keyboard/mouse/audio/video/storage classes	KHAHP-26.04A · KHAHP-26.04B · KHAHP-26.04C
26.05	Rejected HID+extra and RNDIS/network-device patterns	KHAHP-26.05A · KHAHP-26.05B · KHAHP-26.05C
26.06	IPC authorization/notifier behavior	KHAHP-26.06A · KHAHP-26.06B · KHAHP-26.06C
26.07	Hot-plug negative tests and reboot persistence	KHAHP-26.07A · KHAHP-26.07B · KHAHP-26.07C

27 — NetworkManager / Wi-Fi / MAC / DHCP

Observed implementation/evidence: NetworkManager uses a Wi-Fi connection and randomized identity controls were historically configured.

Acceptance boundary / next proof: Over-the-air MAC and DHCP identity comparison on an owned router remains a dedicated test.

Historical action references: `U25-005`, `U25-006`.

Family	Exact technical scope	A/B/C evidence IDs
27.01	NetworkManager package/service state	KHAHP-27.01A · KHAHP-27.01B · KHAHP-27.01C
27.02	Saved Wi-Fi profile inventory and autoconnect	KHAHP-27.02A · KHAHP-27.02B · KHAHP-27.02C
27.03	Active-profile cloned-MAC and DHCP privacy settings	KHAHP-27.03A · KHAHP-27.03B · KHAHP-27.03C
27.04	Scan MAC randomization	KHAHP-27.04A · KHAHP-27.04B · KHAHP-27.04C
27.05	Ethernet MAC randomization policy	KHAHP-27.05A · KHAHP-27.05B · KHAHP-27.05C
27.06	Runtime MAC versus permanent hardware MAC	KHAHP-27.06A · KHAHP-27.06B · KHAHP-27.06C
27.07	DHCP client identifier and profile persistence	KHAHP-27.07A · KHAHP-27.07B · KHAHP-27.07C

28 — IPv6 Policy

Observed implementation/evidence: The inspected kernel boot used `ipv6.disable=1`, and the USER network snapshot showed no IPv6 route.

Acceptance boundary / next proof: Explicitly decide whether to retain disabled IPv6 or move to a separately reviewed dual-stack privacy architecture.

Historical action references: `U25-007`, `U25-019`.

Family	Exact technical scope	A/B/C evidence IDs
28.01	ipv6.disable=1 boot parameter	KHAHP-28.01A · KHAHP-28.01B · KHAHP-28.01C

Family	Exact technical scope	A/B/C evidence IDs
28.02	Absence of IPv6 addresses	KHAHP-28.02A · KHAHP-28.02B · KHAHP-28.02C
28.03	Absence of IPv6 routes	KHAHP-28.03A · KHAHP-28.03B · KHAHP-28.03C
28.04	NetworkManager IPv6 method/state	KHAHP-28.04A · KHAHP-28.04B · KHAHP-28.04C
28.05	Resolver absence of IPv6 nameservers	KHAHP-28.05A · KHAHP-28.05B · KHAHP-28.05C
28.06	IPv6 TCP/UDP listener exposure	KHAHP-28.06A · KHAHP-28.06B · KHAHP-28.06C
28.07	Reboot persistence and DNS/HTTPS regression test	KHAHP-28.07A · KHAHP-28.07B · KHAHP-28.07C

29 — Unbound / Local DNS Architecture

Observed implementation/evidence: Unbound is active and configured, and the USER snapshot showed a `127.0.0.1:53` listener; local DNS resolution was functional.

Acceptance boundary / next proof: Distinguish effective system resolver wiring from application bypass during network transitions.

Historical action references: `U25-008`, `U25-020`.

Family	Exact technical scope	A/B/C evidence IDs
29.01	Unbound package/service state	KHAHP-29.01A · KHAHP-29.01B · KHAHP-29.01C
29.02	127.0.0.1:53 UDP/TCP listener and non-loopback absence	KHAHP-29.02A · KHAHP-29.02B · KHAHP-29.02C
29.03	resolv.conf local-resolver wiring	KHAHP-29.03A · KHAHP-29.03B · KHAHP-29.03C
29.04	NetworkManager effective DNS assignment	KHAHP-29.04A · KHAHP-29.04B · KHAHP-29.04C
29.05	Unbound configuration include tree validation	KHAHP-29.05A · KHAHP-29.05B · KHAHP-29.05C
29.06	DoT forward-zone and IPv6 source policy	KHAHP-29.06A · KHAHP-29.06B · KHAHP-29.06C
29.07	USER/SYSMAINT service behavior and persistence	KHAHP-29.07A · KHAHP-29.07B · KHAHP-29.07C

30 — DNSSEC / DoT / DNS Leak Prevention

Observed implementation/evidence: A DNSSEC-valid response set the AD flag; a deliberately invalid DNSSEC domain returned SERVFAIL. Unbound DoT forwarding and a port-853 process observation were previously reported.

Acceptance boundary / next proof: Certificate-negative transport validation and bounded external UDP/TCP 53 bypass verdicts are not yet signed off.

Historical action references: `U25-009`, `U25-010`, `U25-011`, `U25-021`.

Family	Exact technical scope	A/B/C evidence IDs
30.01	DNSSEC positive validation test	KHAHP-30.01A · KHAHP-30.01B · KHAHP-30.01C
30.02	Broken-DNSSEC negative test	KHAHP-30.02A · KHAHP-30.02B · KHAHP-30.02C
30.03	DoT forwarding configuration and reachability scope	KHAHP-30.03A · KHAHP-30.03B · KHAHP-30.03C
30.04	Plaintext external UDP/53 block	KHAHP-30.04A · KHAHP-30.04B · KHAHP-30.04C
30.05	Plaintext external TCP/53 block	KHAHP-30.05A · KHAHP-30.05B · KHAHP-30.05C
30.06	Local DNS functionality after firewall enforcement	KHAHP-30.06A · KHAHP-30.06B · KHAHP-30.06C
30.07	DNS leak/bypass and reboot-persistence testing	KHAHP-30.07A · KHAHP-30.07B · KHAHP-30.07C

31 — nftables Firewall

Observed implementation/evidence: nftables is active; privileged rules show input and forward default-drop policies, but the OUTPUT base policy permits traffic with explicit direct-DNS restrictions.

Acceptance boundary / next proof: Correlate final effective OUTPUT rule order and UDP/TCP DNS verdicts at an authorized endpoint; verify normalized reboot persistence.

Historical action references: `U25-012`, `U25-013`.

Family	Exact technical scope	A/B/C evidence IDs
31.01	nftables package/service/runtime ruleset state	KHAHP-31.01A · KHAHP-31.01B · KHAHP-31.01C
31.02	Dedicated inet kicksecure_hardening table	KHAHP-31.02A · KHAHP-31.02B · KHAHP-31.02C
31.03	Input default-drop policy	KHAHP-31.03A · KHAHP-31.03B · KHAHP-31.03C
31.04	Forward default-drop policy	KHAHP-31.04A · KHAHP-31.04B · KHAHP-31.04C
31.05	Loopback, established/related and invalid handling	KHAHP-31.05A · KHAHP-31.05B · KHAHP-31.05C
31.06	DHCP/ICMP and required inbound exceptions	KHAHP-31.06A · KHAHP-31.06B · KHAHP-31.06C
31.07	Static syntax, reload safety and reboot persistence	KHAHP-31.07A · KHAHP-31.07B · KHAHP-31.07C

32 — LAN Discovery / Multicast / LLMNR / mDNS

Observed implementation/evidence: LAN discovery service and listener inventory was collected.

Acceptance boundary / next proof: Peer-origin mDNS, LLMNR and SSDP reply-negative tests remain inconclusive without an owned LAN peer.

Historical action references: `U25-014`, `U25-022`.

Family	Exact technical scope	A/B/C evidence IDs
32.01	Avahi/mDNS daemon state	KHAHP-32.01A · KHAHP-32.01B · KHAHP-32.01C
32.02	LLMNR/resolved multicast name-resolution state	KHAHP-32.02A · KHAHP-32.02B · KHAHP-32.02C
32.03	wsdd/Samba discovery state	KHAHP-32.03A · KHAHP-32.03B · KHAHP-32.03C
32.04	rpcbind/minidlna/UPnP service state	KHAHP-32.04A · KHAHP-32.04B · KHAHP-32.04C
32.05	Multicast/discovery TCP+UDP listener inventory	KHAHP-32.05A · KHAHP-32.05B · KHAHP-32.05C
32.06	Firewall exposure of discovery ports	KHAHP-32.06A · KHAHP-32.06B · KHAHP-32.06C
32.07	Negative LAN discovery and reboot-persistence test	KHAHP-32.07A · KHAHP-32.07B · KHAHP-32.07C

33 — Tor / SOCKS / DNS Isolation

Observed implementation/evidence: Tor was active, with a loopback SOCKS listener observed on port 9050.

Acceptance boundary / next proof: Prove routing and fail-closed behavior for only the applications intended to use Tor; do not stop system-wide services casually.

Historical action references: U25-015.

Family	Exact technical scope	A/B/C evidence IDs
33.01	Tor package/service state	KHAHP-33.01A · KHAHP-33.01B · KHAHP-33.01C
33.02	SOCKS listener 127.0.0.1:9050 and non-loopback absence	KHAHP-33.02A · KHAHP-33.02B · KHAHP-33.02C
33.03	Tor configuration provenance and syntax validation	KHAHP-33.03A · KHAHP-33.03B · KHAHP-33.03C
33.04	Application proxy routing where intended	KHAHP-33.04A · KHAHP-33.04B · KHAHP-33.04C
33.05	Tor DNS isolation behavior	KHAHP-33.05A · KHAHP-33.05B · KHAHP-33.05C
33.06	Clearnet bypass classification for Tor-mode applications	KHAHP-33.06A · KHAHP-33.06B · KHAHP-33.06C
33.07	Bootstrap, circuit, restart and failure-mode behavior	KHAHP-33.07A · KHAHP-33.07B · KHAHP-33.07C

34 — sdwdate / Secure Time / Boot Clock

Observed implementation/evidence: sdwdate, boot-time and secure-time components were observed in the historical baseline.

Acceptance boundary / next proof: Use time/log chronology and disposable-VM rollback tests rather than altering the live system clock.

Historical action references: U25-016, U25-023.

Family	Exact technical scope	A/B/C evidence IDs
34.01	sdwdate package/service state	KHAHP-34.01A · KHAHP-34.01B · KHAHP-34.01C
34.02	timesanitycheck execution state	KHAHP-34.02A · KHAHP-34.02B · KHAHP-34.02C
34.03	bootclockrandomization execution state	KHAHP-34.03A · KHAHP-34.03B · KHAHP-34.03C
34.04	System clock sanity metadata	KHAHP-34.04A · KHAHP-34.04B · KHAHP-34.04C
34.05	NTP/systemd-timesyncd conflicts	KHAHP-34.05A · KHAHP-34.05B · KHAHP-34.05C
34.06	USER versus SYSMAINT time-source behavior	KHAHP-34.06A · KHAHP-34.06B · KHAHP-34.06C
34.07	Time rollback/jump detection and log coherence	KHAHP-34.07A · KHAHP-34.07B · KHAHP-34.07C

35 — Entropy / RNG

Observed implementation/evidence: The later USER evidence showed entropy availability and strong kernel randomization settings; RNG services were enabled in the system baseline.

Acceptance boundary / next proof: Cloned-image seed uniqueness is a future release test and not a claim about a released product.

Historical action references: `U35-001`.

Family	Exact technical scope	A/B/C evidence IDs
35.01	Kernel RNG readiness at boot	KHAHP-35.01A · KHAHP-35.01B · KHAHP-35.01C
35.02	haveged package/service state	KHAHP-35.02A · KHAHP-35.02B · KHAHP-35.02C
35.03	jitterentropy-rngd package/service state	KHAHP-35.03A · KHAHP-35.03B · KHAHP-35.03C
35.04	random.trust_cpu boot policy	KHAHP-35.04A · KHAHP-35.04B · KHAHP-35.04C
35.05	random.trust_bootloader boot policy	KHAHP-35.05A · KHAHP-35.05B · KHAHP-35.05C
35.06	Entropy source and /dev/random behavior	KHAHP-35.06A · KHAHP-35.06B · KHAHP-35.06C
35.07	Early-boot entropy persistence and regression evidence	KHAHP-35.07A · KHAHP-35.07B · KHAHP-35.07C

36 — Browser DoH / WebRTC / Telemetry

Observed implementation/evidence: The active Firefox `default-release` configuration was observed with DoH/TRR mode 5, WebRTC disabled and relevant prefetch settings disabled. Enabled extension inventory was empty.

Acceptance boundary / next proof: An effective browser-to-network DNS/DoH bypass test using an authorized endpoint remains distinct from preference-file inspection.

Historical action references: `U35-002`.

Family	Exact technical scope	A/B/C evidence IDs
36.01	Browser DNS-over-HTTPS/TRR setting	KHAHP-36.01A · KHAHP-36.01B · KHAHP-36.01C
36.02	Browser proxy DNS routing behavior	KHAHP-36.02A · KHAHP-36.02B · KHAHP-36.02C
36.03	WebRTC/STUN local/public IP exposure	KHAHP-36.03A · KHAHP-36.03B · KHAHP-36.03C
36.04	Telemetry and crash-report settings	KHAHP-36.04A · KHAHP-36.04B · KHAHP-36.04C
36.05	Safe browsing/prediction/background network features	KHAHP-36.05A · KHAHP-36.05B · KHAHP-36.05C
36.06	Extension/add-on inventory and trust	KHAHP-36.06A · KHAHP-36.06B · KHAHP-36.06C
36.07	Browser update, profile persistence and leak tests	KHAHP-36.07A · KHAHP-36.07B · KHAHP-36.07C

37 — Application / File / Metadata Privacy

Observed implementation/evidence: `org.gnome.desktop.thumbnailers disable-all` changed from false to true, and later USER evidence still read true. PCManFM-Qt thumbnail disabling, a synthetic image test and reboot persistence were observed.

Acceptance boundary / next proof: Third-party application history/thumbnail backends and privacy tradeoffs require explicit acceptance. The earlier apply/verify archive self-hash defect is packaging-only.

Historical action references: `U35-003`.

Family	Exact technical scope	A/B/C evidence IDs
37.01	Desktop search/indexing services	KHAHP-37.01A · KHAHP-37.01B · KHAHP-37.01C
37.02	Thumbnail and preview cache behavior	KHAHP-37.02A · KHAHP-37.02B · KHAHP-37.02C
37.03	Recent-files/history and clipboard retention	KHAHP-37.03A · KHAHP-37.03B · KHAHP-37.03C
37.04	Document/image metadata handling	KHAHP-37.04A · KHAHP-37.04B · KHAHP-37.04C
37.05	Temporary-file and application-cache exposure	KHAHP-37.05A · KHAHP-37.05B · KHAHP-37.05C
37.06	Sandboxed versus unsandboxed application inventory	KHAHP-37.06A · KHAHP-37.06B · KHAHP-37.06C
37.07	Metadata-removal workflow and regression testing	KHAHP-37.07A · KHAHP-37.07B · KHAHP-37.07C

38 — Camera / Microphone / Power / Screen Lock

Observed implementation/evidence: Normal USER opening `/dev/video0`, `/dev/video1` and `/dev/snd/pcmC0D0c` was denied without capture. Bluetooth was rfkill-soft-blocked with inactive service. `swaylock` successfully locked the desktop in a shorter manual test.

Acceptance boundary / next proof: Five-minute unattended lock persistence, lid/resume authentication (if policy requires it), keyboard/mouse hotplug and hibernation decision remain distinct tests.

Historical action references: `U35-004`.

Family	Exact technical scope	A/B/C evidence IDs
38.01	Camera hardware/device exposure	KHAHP-38.01A · KHAHP-38.01B · KHAHP-38.01C
38.02	Microphone hardware/device exposure	KHAHP-38.02A · KHAHP-38.02B · KHAHP-38.02C
38.03	Bluetooth/rfkill interaction with peripherals	KHAHP-38.03A · KHAHP-38.03B · KHAHP-38.03C
38.04	Screen-lock timeout and authentication	KHAHP-38.04A · KHAHP-38.04B · KHAHP-38.04C
38.05	Suspend/lock-on-lid and idle behavior	KHAHP-38.05A · KHAHP-38.05B · KHAHP-38.05C
38.06	Hibernation configuration and encrypted-swap implications	KHAHP-38.06A · KHAHP-38.06B · KHAHP-38.06C
38.07	Shutdown-versus-suspend operational policy	KHAHP-38.07A · KHAHP-38.07B · KHAHP-38.07C

39 — Firmware / BIOS / Secure Boot / TPM

Observed implementation/evidence: OWNER COMPLETED / EXCLUDED FROM RETEST. BIOS/UEFI checks have already been passed according to the owner. Other evidence also reported Secure Boot enabled; no further BIOS testing is requested.

Acceptance boundary / next proof: Preserve the owner completion declaration and dated record. It is not an independent auditor certification or 21 signed A/B/C gates.

Family	Exact technical scope	A/B/C evidence IDs
39.01	BIOS/UEFI vendor, model and firmware version	KHAHP-39.01A · KHAHP-39.01B · KHAHP-39.01C
39.02	fwupd device/security-attribute inventory	KHAHP-39.02A · KHAHP-39.02B · KHAHP-39.02C
39.03	Secure Boot state and Setup Mode	KHAHP-39.03A · KHAHP-39.03B · KHAHP-39.03C
39.04	TPM presence, version and ownership state	KHAHP-39.04A · KHAHP-39.04B · KHAHP-39.04C
39.05	BIOS/UEFI administrator password manual check	KHAHP-39.05A · KHAHP-39.05B · KHAHP-39.05C
39.06	PXE/CSM/external-boot/manual boot- order policy	KHAHP-39.06A · KHAHP-39.06B · KHAHP-39.06C
39.07	Firmware update, DKMS/MOK and Secure-Boot readiness	KHAHP-39.07A · KHAHP-39.07B · KHAHP-39.07C

40 — Auditd / AIDE / Integrity Monitoring

Observed implementation/evidence: The `dailyaidecheck.timer` was enabled and active. In a SYSMAINT snapshot, audit was enabled but `auditctl -s` showed `pid 0` and `auditctl -l` reported no active rules.

Acceptance boundary / next proof: The owner targets 01–40 completion, but a benign audit event and existing AIDE baseline alert lifecycle were not supplied. Do not claim final Section 40 enforcement acceptance yet.

Family	Exact technical scope	A/B/C evidence IDs
40.01	auditd package/service state	KHAHP-40.01A · KHAHP-40.01B · KHAHP-40.01C
40.02	Audit rule inventory and immutable/finalize policy	KHAHP-40.02A · KHAHP-40.02B · KHAHP-40.02C
40.03	Privileged-command and identity- change audit coverage	KHAHP-40.03A · KHAHP-40.03B · KHAHP-40.03C
40.04	AIDE package/database/configuration state	KHAHP-40.04A · KHAHP-40.04B · KHAHP-40.04C
40.05	File-integrity baseline and scheduled checks	KHAHP-40.05A · KHAHP-40.05B · KHAHP-40.05C
40.06	Audit-log ownership, rotation and tamper res- istance	KHAHP-40.06A · KHAHP-40.06B · KHAHP-40.06C
40.07	CIS/DISA accountability-gap analysis	KHAHP-40.07A · KHAHP-40.07B · KHAHP-40.07C

41 — Cron / Timers / Persistence / Supply Chain

Observed implementation/evidence: Scheduled jobs, startup entries and software supply-chain attribution are in the planned final assurance phase.

Acceptance boundary / next proof: Map provenance, signatures and cross-reboot persistence; future work.

Family	Exact technical scope	A/B/C evidence IDs
41.01	cron/anacron job inventory	KHAHP-41.01A · KHAHP-41.01B · KHAHP-41.01C
41.02	systemd timer inventory	KHAHP-41.02A · KHAHP-41.02B · KHAHP-41.02C
41.03	User autostart and desktop persistence invent- ory	KHAHP-41.03A · KHAHP-41.03B · KHAHP-41.03C
41.04	Package repository and signing-key provenance	KHAHP-41.04A · KHAHP-41.04B · KHAHP-41.04C
41.05	APT pinning/holds and update-policy review	KHAHP-41.05A · KHAHP-41.05B · KHAHP-41.05C
41.06	Unexpected startup scripts/services/modules	KHAHP-41.06A · KHAHP-41.06B · KHAHP-41.06C
41.07	Supply-chain drift and reboot- persistence evid- ence	KHAHP-41.07A · KHAHP-41.07B · KHAHP-41.07C

42 — Credentials / SSH Keys / Secrets / Backup

Observed implementation/evidence: Credentials and backup planning are documented; a private production-grade isolated restore has not been shown.

Acceptance boundary / next proof: Approve encrypted backups, retention, custody and isolated restoration.

Family	Exact technical scope	A/B/C evidence IDs
42.01	SSH private-key inventory and permissions	KHAHP-42.01A · KHAHP-42.01B · KHAHP-42.01C
42.02	API/token/secret storage locations	KHAHP-42.02A · KHAHP-42.02B · KHAHP-42.02C

Family	Exact technical scope	A/B/C evidence IDs
42.03	Password-manager selection and vault policy	KHAHP-42.03A · KHAHP-42.03B · KHAHP-42.03C
42.04	FIDO2/YubiKey readiness and hardware-backed keys	KHAHP-42.04A · KHAHP-42.04B · KHAHP-42.04C
42.05	Recovery-code generation and offline storage	KHAHP-42.05A · KHAHP-42.05B · KHAHP-42.05C
42.06	Encrypted backup creation and retention	KHAHP-42.06A · KHAHP-42.06B · KHAHP-42.06C
42.07	Restore test, key recovery and disaster-recovery evidence	KHAHP-42.07A · KHAHP-42.07B · KHAHP-42.07C

43 — CIS / DISA GPOS / KSPP Mapping

Observed implementation/evidence: CIS Debian 13 Level 2, DISA GPOS/SRG and KSPP have been identified as comparison frameworks. The project also identifies native Kicksecure protections and extra engineering requirements that may extend conventional baseline scope.

Acceptance boundary / next proof: Build a control-by-control mapping of each applicable benchmark requirement, observed inherited safeguard, project extension and justified exception. The comparison must be evidence-backed before claiming measured superiority over a specific bank baseline, CIS score or official certification.

Family	Exact technical scope	A/B/C evidence IDs
43.01	CIS Debian 13 Level 2 applicability mapping	KHAHP-43.01A · KHAHP-43.01B · KHAHP-43.01C
43.02	DISA GPOS/SRG objective mapping	KHAHP-43.02A · KHAHP-43.02B · KHAHP-43.02C
43.03	KSPP/kernel-self-protection mapping	KHAHP-43.03A · KHAHP-43.03B · KHAHP-43.03C
43.04	Kicksecure-native equivalent control mapping	KHAHP-43.04A · KHAHP-43.04B · KHAHP-43.04C
43.05	Project-extension control mapping	KHAHP-43.05A · KHAHP-43.05B · KHAHP-43.05C
43.06	Exception/N-A/manual-control rationale	KHAHP-43.06A · KHAHP-43.06B · KHAHP-43.06C
43.07	Evidence citation and no-certification disclaimer	KHAHP-43.07A · KHAHP-43.07B · KHAHP-43.07C

44 — Final Acceptance / Evidence / Scoring

Observed implementation/evidence: A/B/C acceptance methodology is defined for 308 families and 924 gates.

Acceptance boundary / next proof: Final owner/reviewer signoff, negative testing, recoverability and evidence completeness.

Family	Exact technical scope	A/B/C evidence IDs
44.01	Full reboot persistence acceptance	KHAHP-44.01A · KHAHP-44.01B · KHAHP-44.01C
44.02	USER-mode regression acceptance	KHAHP-44.02A · KHAHP-44.02B · KHAHP-44.02C

Family	Exact technical scope	A/B/C evidence IDs
44.03	SYSMAINT-mode regression acceptance	KHAHP-44.03A · KHAHP-44.03B · KHAHP-44.03C
44.04	Network/DNS/firewall/Tor/USB negative-test acceptance	KHAHP-44.04A · KHAHP-44.04B · KHAHP-44.04C
44.05	Recovery/rollback/backup acceptance	KHAHP-44.05A · KHAHP-44.05B · KHAHP-44.05C
44.06	Evidence completeness, hashes and chain-of-custody bundle	KHAHP-44.06A · KHAHP-44.06B · KHAHP-44.06C
44.07	Final scorecard, residual risk and operator sign-off	KHAHP-44.07A · KHAHP-44.07B · KHAHP-44.07C

6. Historical 01–38 action reconciliation — all 63 IDs

These 63 inherited action records are carried forward to preserve **hardening opportunities, configuration concerns, owner decisions and evidence gaps** that can materially improve security. They are part of the 69-item mixed historical backlog, not a claim of 63 independently proven exploits. Later successful demonstrations are retained as engineering results; remaining requirements are explicitly shown and **do not become signed acceptance without review**. Section 39 BIOS is not retested.

ID	Sec.	Engineering issue	Evidence disposition as of 18 Oct 2026
U01-001	7	/tmp and /dev/shm noexec compatibility decision	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-002	7	/boot/efi-before-/boot fstab ordering review	CONFIGURATION FIX VERIFIED (18 OCT); NEXT-BOOT PERSISTENCE NOT YET ATTESTED
U01-003	2	permission-hardener package/provenance classifier	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-004	5	sync account shell classification	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-005	12	Unverified kernel tokens / CFI-KCFI claims	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-006	13	vesafb module applicability / dry-run review	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-007	14	USER versus SYSMAINT DKMS evidence reconciliation	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-008	8	Physical GRUB edit-denial test	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-009	6	LUKS key recovery and backup restore	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-010	10	Firmware/preboot DMA protection acceptance	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-011	7	Post-reboot SYSMAINT persistence evidence	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-012	7	/home, /var/tmp and noexec compatibility review	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U01-013	14	Signed-module negative test and update regression	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED

ID	Sec.	Engineering issue	Evidence disposition as of 18 Oct 2026
U01-014	7	/tmp nodev,nosuid hardening — implemented baseline	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-001	15	Strict Firefox AppArmor confinement is not established	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-002	15	AppArmor service policy and denial regression	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-003	15	USER versus SYSMAINT browser-label persistence	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-004	16	Privileged binary provenance/drift review	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-005	16	gst-ptp-helper capability policy/ compatibility	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-006	17	setup-dist status-files 0777 exception monitoring	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-007	17	Sensitive directory/ACL/umask and tmpfs drift	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-008	18	faillock recovery and self-lockout policy	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-009	18	Passphrase/aging policy ownership	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-010	18	PAM USER/SYSMAINT interaction regression	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-011	19	Network SSH masking and listener persistence	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-012	19	Local SSH AF_UNIX socket and client key policy	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-013	20	Role-specific systemd service drift and listening sockets	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-014	20	Classify unexpected/unused services safely	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-015	21	Unbound systemd exposure score needs design decision	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-016	21	Service sandbox capability regression	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-017	22	Core dump/pstore suppression drift	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-018	22	Crash-forensics versus privacy trade-off	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-019	23	Journal retention, permissions and rotation policy	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-020	23	sdwdate time and journal chronology after reboot	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-021	24	Owner-approved panic/recovery policy and escape paths	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U15-022	24	Emergency shutdown and controlled recovery evidence	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED

ID	Sec.	Engineering issue	Evidence disposition as of 18 Oct 2026
U25-001	26	External USB keyboard possibly allowed	POLICY CORRECTED: EXTERNAL KEYBOARDS ALLOWED; FUNCTIONAL TEST TO RECONCILE
U25-002	26	Composite/USB network rejection and persistence	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-003	25	Swap crypt mapping and shutdown key lifecycle	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-004	25	RAM wipe shutdown execution and hardware limits	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-005	27	Actual Wi-Fi MAC identity	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-006	27	DHCP identity rotation	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-007	28	Alternate IPv6 network profiles	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-008	29	Unbound response to DHCP renew/network loss	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-009	30	Unbound-owned authenticated DoT	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-010	30	Direct UDP/TCP 53 egress	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-011	30	Active browser DNS/DoH preferences	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-012	31	Effective firewall output chain	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-013	31	Exact loaded-ruleset persistence	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-014	32	Peer-side discovery proof	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-015	33	Tor Browser routing/DNS/fail-close	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-016	34	Secure-time unit chronology	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-017	25	Original implementation provenance	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-018	26	USB boot media usability versus detection	RECOVERY MEDIA USB AUTHORIZED/DETECTED; ACCESSIBILITY/INSTALL MODE SEPARATE
U25-019	28	IPv6 goal mismatch / explicit policy	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-020	29	USER/SYSMAINT service role semantics	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U25-021	30	DNSSEC positive/negative evidence signoff	DNSSEC POSITIVE/NEGATIVE OBSERVED; FORMAL CLAIM REVIEW OPEN
U25-022	32	Actual scope of local discovery services	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED

ID	Sec.	Engineering issue	Evidence disposition as of 18 Oct 2026
U25-023	34	Production clock rollback exception	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U35-001	35	Early-boot entropy and random-seed lifecycle	REVIEW / PROOF OR OWNER DECISION NOT YET SIGNED
U35-002	36	Firefox effective DoH, WebRTC and telemetry behavior	FIREFOX SETTINGS VERIFIED; EXTERNAL BROWSER PATH NEGATIVE STILL OPEN
U35-003	37	Application history, thumbnails, metadata and confinement regression	THUMBNAILS AND REBOOT TEST VERIFIED; OTHER APP BEHAVIOR NOT FULLY SIGNED
U35-004	38	Camera/microphone hotplug and real lock/lid-resume enforcement	CAMERA/MIC USER DENIAL + BT BLOCK + SHORT LOCK PASS; 5-MIN/LID OPEN

Essential reconciliation decisions

1. **Owner device policy prevails:** external keyboards, two approved mice and USB removable/recovery storage are allowed. Older testing notes requesting keyboard rejection are superseded.
2. **Confirmed changes are not identical to confirmed exploit findings.** Track pre-state, root cause, authorization, exact delta, positive and negative post-state before classifying a true fixed weakness.
3. **Section 07 boot ordering:** immediate root verification passed after correction; avoid reapplying the edit. Preserve backup and collect persistence only when a normal owner reboot occurs.
4. **Section 15 Firefox:** the vendor unconfined placeholder remains an explicitly open MAC enhancement, even though Firefox subprocess seccomp filters exist.
5. **Section 40 audit events:** a working AIDE timer does not prove event coverage. The latest SYSMAINT run observed no active audit rules and PID 0; troubleshoot before a full assurance claim.
6. **Section 39 BIOS:** prior owner completion is accepted as the project process decision and excluded from future retesting; it does not forge formal independent sign-off.

7. Commercial image engineering and supply-chain acceptance

Proposed KHAHP Premium Hardened OS target price: **USD $1,250**; intended distributions are physical ISO/USB, KVM/QCOW2, and VirtualBox/OVA. No public downloadable commercial image or certification is asserted. A clean build must never clone the personal development machine, SSH secrets, random seeds or machine identity.

Clean-image common requirements

Planned release ID	Required demonstration	Current program state
KHAHP-REL-COM-01	Reproducible upstream source	PROPOSED · NOT TESTED
KHAHP-REL-COM-02	Package provenance	PROPOSED · NOT TESTED
KHAHP-REL-COM-03	Build isolation	PROPOSED · NOT TESTED
KHAHP-REL-COM-04	Credential exclusion	PROPOSED · NOT TESTED
KHAHP-REL-COM-05	Unique machine identity	PROPOSED · NOT TESTED
KHAHP-REL-COM-06	Unique SSH host keys	PROPOSED · NOT TESTED
KHAHP-REL-COM-07	SSH access policy	PROPOSED · NOT TESTED

Planned release ID	Required demonstration	Current program state
KHAHP-REL-COM-08	Privileges and boot-role separation	PROPOSED · NOT TESTED
KHAHP-REL-COM-09	Default firewall	PROPOSED · NOT TESTED
KHAHP-REL-COM-10	DNS resolver protection	PROPOSED · NOT TESTED
KHAHP-REL-COM-11	Network privacy behavior	PROPOSED · NOT TESTED
KHAHP-REL-COM-12	MAC and device privacy	PROPOSED · NOT TESTED
KHAHP-REL-COM-13	Mandatory access controls	PROPOSED · NOT TESTED
KHAHP-REL-COM-14	Kernel hardening	PROPOSED · NOT TESTED
KHAHP-REL-COM-15	Audit visibility	PROPOSED · NOT TESTED
KHAHP-REL-COM-16	File integrity monitoring	PROPOSED · NOT TESTED
KHAHP-REL-COM-17	Upgrade safety	PROPOSED · NOT TESTED
KHAHP-REL-COM-18	Backup and restoration	PROPOSED · NOT TESTED
KHAHP-REL-COM-19	Release signing	PROPOSED · NOT TESTED
KHAHP-REL-COM-20	Customer documentation and licensing	PROPOSED · NOT TESTED

Physical ISO / USB requirements

Planned release ID	Required demonstration	Current program state
KHAHP-REL-ISO-01	UEFI USB boot	PROPOSED · NOT TESTED
KHAHP-REL-ISO-02	Installer storage detection	PROPOSED · NOT TESTED
KHAHP-REL-ISO-03	Fresh encrypted installation	PROPOSED · NOT TESTED
KHAHP-REL-ISO-04	Encryption recovery	PROPOSED · NOT TESTED
KHAHP-REL-ISO-05	Secure Boot enrollment	PROPOSED · NOT TESTED
KHAHP-REL-ISO-06	Firmware Setup Mode	PROPOSED · NOT TESTED
KHAHP-REL-ISO-07	TPM optional path	PROPOSED · NOT TESTED
KHAHP-REL-ISO-08	Intel and AMD support	PROPOSED · NOT TESTED
KHAHP-REL-ISO-09	Graphics and desktop	PROPOSED · NOT TESTED
KHAHP-REL-ISO-10	Wireless networking	PROPOSED · NOT TESTED
KHAHP-REL-ISO-11	Ethernet and hotplug	PROPOSED · NOT TESTED
KHAHP-REL-ISO-12	USBGuard defaults	PROPOSED · NOT TESTED
KHAHP-REL-ISO-13	Installer device safety	PROPOSED · NOT TESTED
KHAHP-REL-ISO-14	DMA/IOMMU capabilities	PROPOSED · NOT TESTED
KHAHP-REL-ISO-15	Signed module lifecycle	PROPOSED · NOT TESTED
KHAHP-REL-ISO-16	Physical webcam/microphone	PROPOSED · NOT TESTED
KHAHP-REL-ISO-17	Power and RAM-wipe process	PROPOSED · NOT TESTED
KHAHP-REL-ISO-18	BIOS recovery restrictions	PROPOSED · NOT TESTED
KHAHP-REL-ISO-19	Multi-boot/firmware resilience	PROPOSED · NOT TESTED

Planned release ID	Required demonstration	Current program state
KHAHP-REL-ISO-20	Fresh-install two-device comparison	PROPOSED · NOT TESTED

KVM / QCOW2 requirements

Planned release ID	Required demonstration	Current program state
KHAHP-REL-KVM-01	QCOW2 import and boot	PROPOSED · NOT TESTED
KHAHP-REL-KVM-02	Fresh clone identity	PROPOSED · NOT TESTED
KHAHP-REL-KVM-03	Virtual firmware choice	PROPOSED · NOT TESTED
KHAHP-REL-KVM-04	Virtual TPM support	PROPOSED · NOT TESTED
KHAHP-REL-KVM-05	Host-guest firewall isolation	PROPOSED · NOT TESTED
KHAHP-REL-KVM-06	Guest disk encryption	PROPOSED · NOT TESTED
KHAHP-REL-KVM-07	Virtio device support	PROPOSED · NOT TESTED
KHAHP-REL-KVM-08	KVM snapshot and restore	PROPOSED · NOT TESTED
KHAHP-REL-KVM-09	KVM version regression	PROPOSED · NOT TESTED
KHAHP-REL-KVM-10	QCOW2 sanitized release	PROPOSED · NOT TESTED

VirtualBox / OVA requirements

Planned release ID	Required demonstration	Current program state
KHAHP-REL-OVA-01	OVA packaging and import	PROPOSED · NOT TESTED
KHAHP-REL-OVA-02	Unique identity on import	PROPOSED · NOT TESTED
KHAHP-REL-OVA-03	Clipboard and drag-drop	PROPOSED · NOT TESTED
KHAHP-REL-OVA-04	Shared-folder isolation	PROPOSED · NOT TESTED
KHAHP-REL-OVA-05	USB passthrough restriction	PROPOSED · NOT TESTED
KHAHP-REL-OVA-06	VirtualBox networking	PROPOSED · NOT TESTED
KHAHP-REL-OVA-07	Guest addition safety	PROPOSED · NOT TESTED
KHAHP-REL-OVA-08	Virtual disk recovery	PROPOSED · NOT TESTED
KHAHP-REL-OVA-09	Supported host matrix	PROPOSED · NOT TESTED
KHAHP-REL-OVA-10	OVA clean release gate	PROPOSED · NOT TESTED

8. Planned completion of Sections 41–44

Section	Required work	Release proof
41 — Persistence / supply chain	Cron/timer/autostart classification, signing and package origin, two-boot drift	Attributed, bounded and stable scheduled activity
42 — Credentials / backup / recovery	Secret inventory without disclosure; owner-approved encrypted backups; isolated restore	Successful recovery and sample hash comparison; custody controls
43 — CIS/DISA/KSPP applicability	Map every relevant requirement, native equivalent and justified N/A	Evidence-linked applicability map; no official certification claim

Section	Required work	Release proof
44 — Final A/B/C sign-off	Independent review, complete negative/regression and rollback proof, dated acceptance	308 families × A/B/C accepted with documented risk exceptions

9. Compliance, controlled change and public disclosure

Execution roles: USER runs normal browser and read-only checks without sudo. SYSMAINT is a separate privileged boot, with independently verified source digests and a root-owned copy for execution. Privileged scripts are never executed in place from a USER-writable transport path.

Change gate: evidence → verified root cause → minimal diff → backup/recovery → owner approval → apply → immediate postflight → bounded negative test → relevant reboot/update persistence → dated A/B/C acceptance.

Publication safety: No hostnames, private identifiers, IP/MAC addresses, raw firewall and browser history, SSH material, real encryption IDs/keys, authentication secrets or raw confidential reports. This register publishes redacted outcomes; underlying logs remain in private custody. Historical Section 37 self-hashing manifest errors must be identified rather than presenting those archives as fully hash-valid.

Reference frameworks: CIS Debian 13 Level 2 Workstation, DISA General Purpose OS SRG and relevant STIG-inspired objectives, KSPP, Debian and Kicksecure security practices. They guide an intentionally deeper hardening and assurance program; Section 43 will document applicable controls, project extensions and measured gaps. They are not third-party compliance certifications.

Public challenge submissions: provide machine-independent provenance, threat model, safely reproducible tests, intended/observed outcomes, minimized changes, rollback proof, and limitations. Do not perform unauthorized external testing or disclose user-specific security details.

10. Evidence source ledger for this revision

v4.1 editorial lineage: Built directly from the owner-uploaded public v4.0 technical register and October 18 evidence ledger. All 44 named sections, 308 numbered families, 924 gate IDs, 63 historical action records and 60 proposed release tests are retained. No device scan or post-October-18 control implementation is implied.

- Pre-existing public register v3.0 (17 October 2026), 308 technical family labels and 60 planned release tests.
- Private KHAHP master v2.0 and 924-gate proof-claims CSV dated 18 October 2026 (all historical signatures OPEN).
- KHAHP compliance standard v3.0, field manual v2.0 and 63-item historical 01–38 closeout list.
- USER 01–38 read-only observations of 18 October 2026; SYSMAINT 65-check archive with validated hashes and zero changes.
- Later USER direct camera/mic open-denial, USBGuard inventory, Firefox process-label/prefs, DNSSEC, AIDE timer and manual short-lock observations.
- Operator-shared SYSMAINT terminal evidence of `/etc/fstab` boot-order correction and clean `findmnt --verify --verbose` validation.
- Section 39 BIOS status: owner declaration, no new test requested.

Evidence limitation: No complete Section 40 audit-event/AIDE alert lifecycle report or formally signed 924-gate acceptance register was supplied in the cited October 18 evidence. This editorial update emphasizes security engineering depth and hardening opportunities without inventing test passes, certification, or more recent machine evidence.